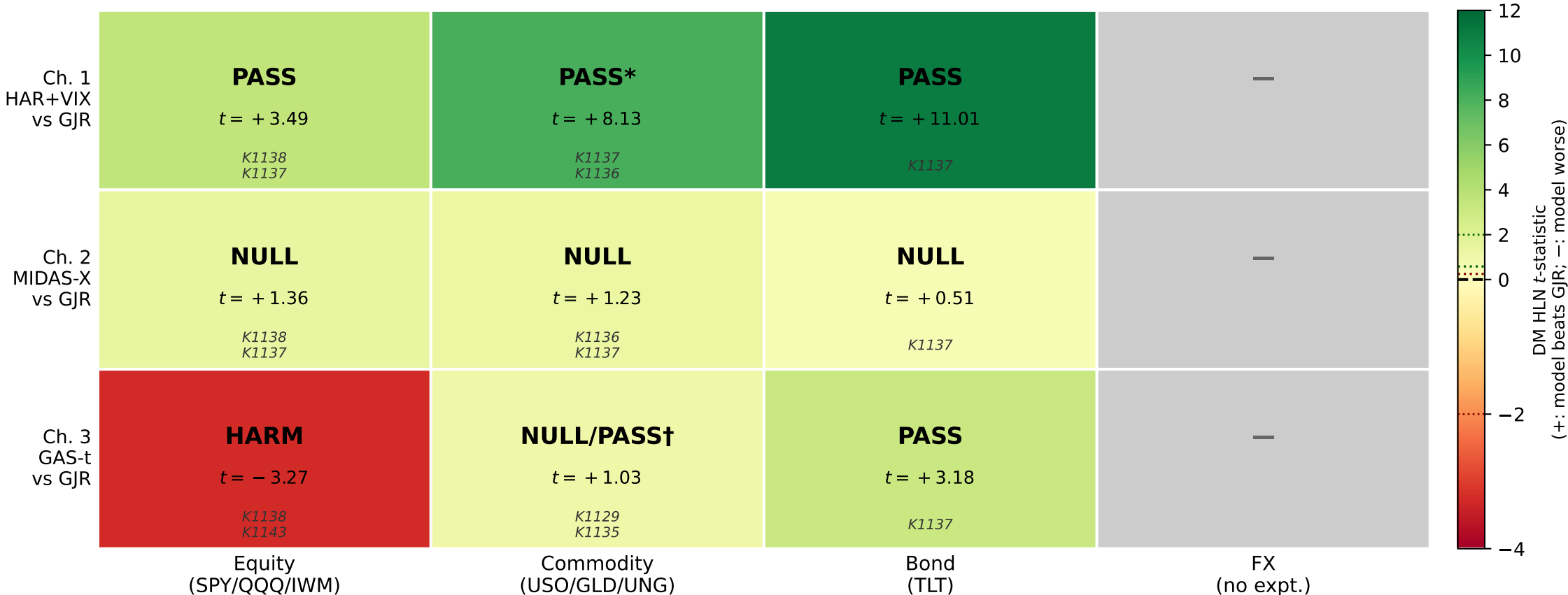


Paper 4: Channel-Specific VIX Sufficiency — Model vs GJR-GARCH Baseline

3 × 4 Out-of-Sample DM HLN t -statistics (OOS: 2020–2026)



PASS: DM $t > +2$, BH $p < 0.05$ — NULL: $|t| \leq 2$ or BH $p \geq 0.05$ — HARM: DM $t < -2$, BH $p < 0.05$
 PASS*: HAR structure beats GJR; VIX marginal NULL (K1136 Fair Test 2) — PASS†: VaR/ES only (skew-t GAS), QLIKE NULL (K1135)